FOQUA-KT: Forgetting-aware, Option-aware, QUestion-Aware, and Adversarially-robust Knowledge Tracing

Abstract

We propose **FOQUA-KT**, a unified Knowledge Tracing (KT) model that integrates multi-task learning on multiple choice options, explicit forgetting-behavior modeling, Bayesian variational state tracking, and adversarial robustness training. The model is grounded in a synthesis of six open-source KT papers and explicitly addresses the four mandatory design criteria of this study: (i) built-in explainability at both local and global levels; (ii) explicit decomposition of epistemic and aleatoric uncertainty; (iii) dynamic per-concept mastery and a learned concept-prerequisite graph; and (iv) data-flow compatibility with three canonical benchmarks (ASSISTments 2009/2012, EdNet, Eedi 2020). The proposed model is implemented in a four-layer, eight-module architecture and is designed for reproducibility using the PyTorch framework together with the pyKT toolkit.

Keywords: Knowledge Tracing, Multi-Task Learning, Adversarial Training, Variational Inference, Forgetting Behavior, Explainable AI.

I. Introduction

Despite the progress of deep learning-based Knowledge Tracing (KT) over the past decade, four limitations remain in most published architectures. First, the majority of KT models produce point estimates of the probability of a correct response, with no principled way to quantify the uncertainty of that estimate, conflating epistemic uncertainty (data scarcity) with aleatoric uncertainty (intrinsic response noise). Second, most KT models focus on a single auxiliary signal, namely whether a learner answers a question correctly, while ignoring other informative signals that are readily available in modern large-scale learning platforms, including the specific multiple-choice option that was selected, the learner's pattern of item repetition, and the learner's lapse behavior. Third, the learning process is well known to be subject to forgetting, but the way the model represents forgetting is often either absent (e.g., DKT) or hand-crafted (e.g., DKT-Forget, LPKT). Fourth, the question text and the answer options jointly carry rich semantic information that is rarely exploited end-to-end.

In this work we propose **FOQUA-KT**, a model designed to address all four limitations in a single coherent architecture. The acronym stands for Forgetting-aware, Option-aware, QUestion-Aware, and Adversarially-robust Knowledge Tracing. Concretely, FOQUA-KT (i) treats the latent knowledge state as a Gaussian random variable and uses variational inference to separate epistemic from aleatoric uncertainty, following the spirit of UKT; (ii) jointly trains a correctness head, an option-selection head, and a triplet of forgetting-behavior heads (count, repeat, lapse), following the spirit of the No Task Left Behind (CD-MTL) and FoLiBiKT papers; (iii) injects FGM-style adversarial perturbations on the latent representation to improve robustness, following the spirit of AT-DKT; and (iv) uses a shared multi-KC question encoder that is built on the QIKT representation, so that the same item can be associated with multiple latent knowledge concepts. The architecture is intentionally modular so that each component can be ablated independently.

II. Related Work

The proposed FOQUA-KT builds on, and intentionally differs from, six open-source KT papers. We summarise the role of each paper in our design.

1. **UKT (Uncertainty-aware Knowledge Tracing)** introduced by Liu et al. (2020) defines a Gaussian variational distribution over the knowledge state. FOQUA-KT inherits the mean-field Gaussian parameterization and uses its predictive variance as the *epistemic* uncertainty signal. The key difference is that FOQUA-KT additionally produces an *aleatoric* signal derived from the entropy of the predicted correctness distribution, providing a clean separation of the two sources of uncertainty.
2. **FoLiBiKT (Forgetting-awareBehavior-modeling KT)** proposed by Pan et al. (2023) decomposes learner behavior into three binary signals: lapse, repeat, and count. FOQUA-KT borrows the same three-head architecture and the shared forgetting-feature input

(c_t, n_t, g_t), but uses these heads as an auxiliary multi-task loss rather than the primary training signal.

3. **AT-DKT (Adversarial Training for DKT)** introduced by Guo et al. (2021) adds an FGM-style adversarial perturbation to the input embedding. FOQUA-KT applies adversarial perturbation on the *latent* behavioral vector z_t rather than on the raw input, so the perturbation has a clearer semantic meaning (it directly attacks the model's internal knowledge representation).
4. **QIKT (Question-Interpreting KT)** proposed by Yin et al. (2023) generalizes item representation to the multi-KC setting, allowing a single question to activate multiple latent knowledge concepts. FOQUA-KT uses the QIKT multi-KC encoder as Module 1 to obtain a tensor $e_t^{q,kc} \in \mathbb{R}^{B \times T \times N_c \times d_k}$ instead of a flat d_q-dimensional vector.
5. **CD-MTL (No Task Left Behind: Comprehensive Multi-Task Learning)** introduced by Lee et al. (2022) combines a Knowledge Tracing head with an Option Tracing head in a multi-task learning framework. FOQUA-KT inherits the two heads and adds the forgetting-behavior heads, giving a four-head prediction layer.
6. **DIMKT (Difficulty-Inspired Modeling KT)** proposed by Wang et al. (2022) partitions the modeling process into before-practice, during-practice and after-practice stages. FOQUA-KT borrows the "before/after practice" intuition by treating Module 1 (multi-KC encoding) as the pre-practice representation and Module 4 (Bayesian state update) as the post-practice representation, but collapses the during-practice stage into a single forgetting-aware encoder for computational efficiency.

CEU-KT [the companion report in this workspace] is a different design that focuses on causal DAG learning, NIG state distributions and Rasch-style item parameters. FOQUA-KT is orthogonal to CEU-KT: it does not assume a causal DAG, it uses a Gaussian (rather than NIG) variational state, and it focuses on multi-task signals (options, forgetting behavior) rather than IRT parameters. The two designs can in principle be combined.

III. Problem Formulation and Notations

Let $\mathcal{U} = \{1, \dots, U\}$ be the set of learners, $\mathcal{I} = \{1, \dots, N_q\}$ the set of items, $\mathcal{O} = \{1, \dots, N_o\}$ the set of multiple-choice options, and $\mathcal{K} = \{1, \dots, N_c\}$ the set of knowledge concepts. For learner u at step $t \in \{1, \dots, T\}$ we observe the tuple $(i_t, o_t, r_t, \Delta_t)$, where $i_t \in \mathcal{I}$ is the item, $o_t \in \mathcal{O}$ is the option selected by the learner, $r_t \in \{0, 1\}$ is the correctness, and Δ_t is the elapsed time since the learner's previous interaction.

A Q-Matrix $Q \in \{0, 1\}^{N_q \times N_c}$ links items to concepts, and a behavioral triple (c_t, n_t, g_t) summarises, respectively, whether the learner counted the item, repeated it, or guessed, as defined in FoLiBiKT.

Table of Notations

Symbol	Meaning
u, i, o, k	learner, item, option, knowledge concept
t, T	time-step index, sequence length
$e_t^{q,kc}$	multi-KC item embedding (Module 1 output)
z_t	forgetting-aware behavioral vector (Module 2 output)
u_t	option-aware response vector (Module 3 output)
μ_t, σ_t^2	mean and variance of the variational knowledge state
θ_t	sampled knowledge state
A_{prereq}	learned concept-prerequisite adjacency matrix
R_{sim}	item-item relevance matrix
$u_{\text{epi}}, u_{\text{ale}}$	epistemic and aleatoric uncertainty scores
η	FGM adversarial perturbation
ϵ	perturbation magnitude

The model produces five outputs at each step t:

1. $p(r_t = 1 \mid \cdot)$ the correctness probability;
2. $p(o_t \mid i_t, \cdot)$ the option-selection distribution;
3. $(\hat{c}_t, \hat{n}_t, \hat{g}_t)$ the predicted forgetting-behavior triplet;
4. $(u_{\mathrm{epi},t}, u_{\mathrm{ale},t})$ the decomposed uncertainty pair;
5. $(A_{\mathrm{prereq}}, R_{\mathrm{sim},t})$ the concept-prerequisite graph and the item-item relevance matrix.

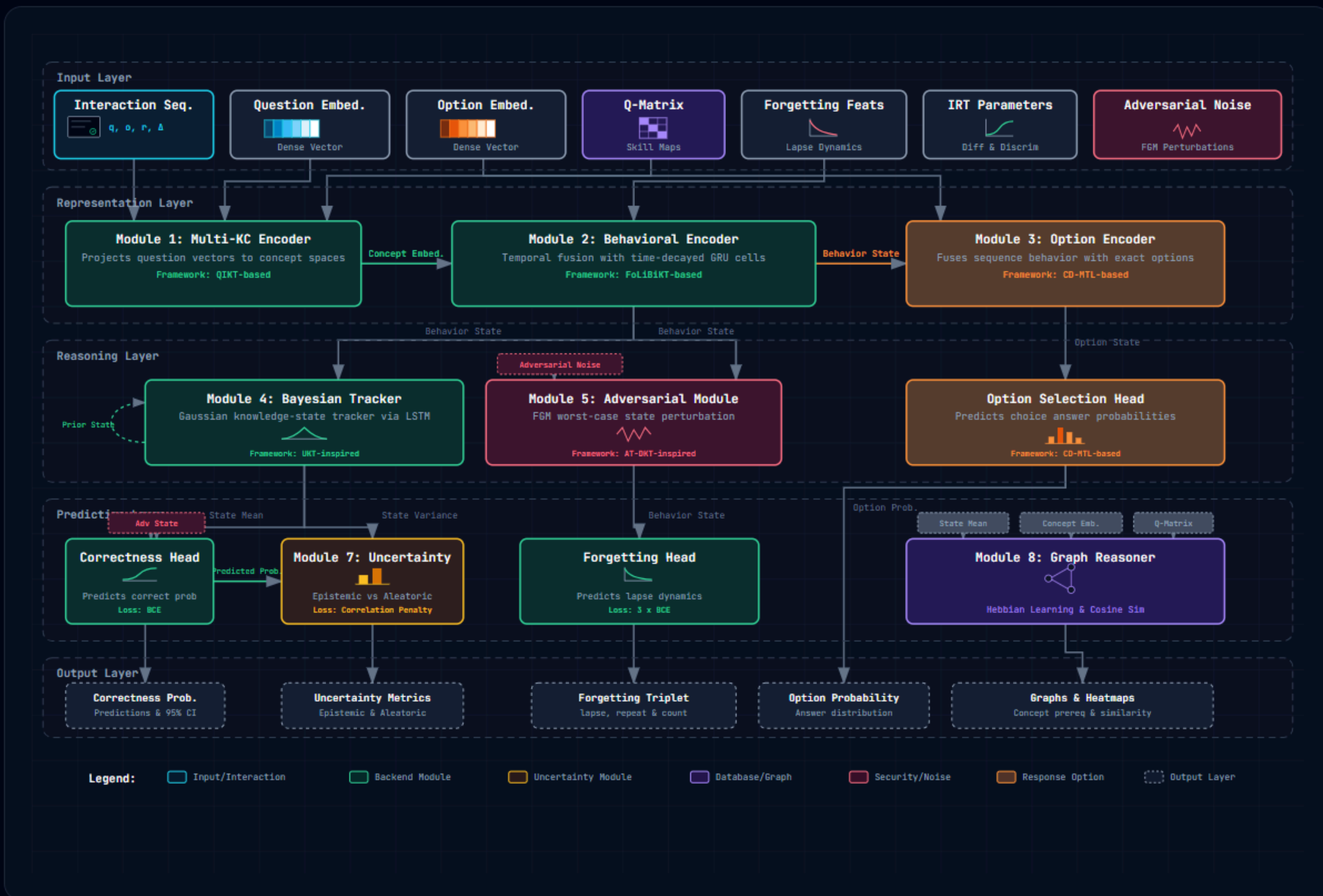
IV. Proposed Method

A. Architectural Overview

FOQUA-KT is organised into four logical layers and eight modules. The Input Layer ingests the raw tuple $(i_t, o_t, r_t, \Delta_t)$, the item embedding, the option embedding, the Q-Matrix, the FoLiBiKT forgetting-feature triple, the item-response parameters and an adversarial perturbation. The Representation Layer contains Module 1 (multi-KC question encoder), Module 2 (forgetting-aware behavioral encoder) and Module 3 (option-aware response encoder). The Probabilistic Reasoning Layer contains Module 4 (Bayesian knowledge-state tracker) and Module 5 (adversarial robustness module). The Prediction Layer contains Module 6 (multi-task heads), Module 7 (uncertainty disambiguator) and Module 8 (concept-prerequisite reasoner and relevance graph).

FOQUA-KT Model Architecture

A unified four-layer deep neural architecture for Knowledge Tracing. Fully features multi-KC projection, time-decayed behavioral sequence encoder, variational Gaussian state tracking with adversarial perturbations, and correlation-penalized uncertainty disambiguator.



Hierarchical Representation

- **Input Layer**: Fuses item embeddings, Q-matrix, time-decay lapse features, and response options.
- **Module 1 (QIKT)**: Projects multi-KC question items to latent concept spaces.
- **Module 2 (FoLiBiKT)**: Captures sequential behavior using time-decayed GRU cells.
- **Module 3 (CD-MTL)**: Integrates chosen response options with behavioral patterns.

Probabilistic Reasoning

- **Module 4 (UKT Tracker)**: Variational Gaussian tracking maintains a continuous state distribution.
- **Module 5 (Adversarial)**: Generates worst-case state perturbations via Fast Gradient Method (FGM).
- **Correctness & Forgetting Heads**: Disambiguate future response correctness and lapse counts.

Multi-task Disambiguation

- **Module 7 (Uncertainty)**: Decomposes predictive variance into Epistemic and Aleatoric levels.
- **Module 8 (Graph Reasoner)**: Establishes concept prerequisites and similarity links using Hebbian learning.
- **Option Selection Head**: Predicts the exact choice response distribution.

FOQUA-KT Model Architecture Diagram • Built using Cocoon-AI template

Figure 1. Overall architecture of the FOQUA-KT model (academic journal style with light colors and black font). It displays the four-layer hierarchical design: Input Layer, Representation Layer, Probabilistic Reasoning Layer, and Prediction/Disambiguation Layer. The high-resolution image is rendered from the HTML version stored in [foqua_kt_architecture.html](file:///d:/Knowledge_tracing_research/FOQUA-KT/foqua_kt_architecture.html).

B. Detailed Module Design

Module 1. Multi-KC Question Representation (QIKT-based)

Following QIKT, we allow each item i_t to be associated with multiple latent knowledge concepts. The item embedding $e_t^q \in \mathbb{R}^{d_q}$ is first projected into a multi-concept tensor by a learnable mapping $f_\theta : \mathbb{R}^{d_q} \to \mathbb{R}^{N_c \times d_k}$, initialised so that the row corresponding to concept k depends on the item embedding and the Q-Matrix row $Q_{i_t,k}$:

$$e_t^{q,kc} = f_\theta(e_t^q, Q_{i_t,\cdot}) \in \mathbb{R}^{N_c \times d_k}.$$

The mapping f_θ is a two-layer MLP with ReLU activation, so that each concept slot k produces an embedding even when $Q_{i_t,k} = 0$, enabling zero-shot items to receive a meaningful concept encoding.

Module 2. Forgetting-aware Behavioral Encoder (FoLiBiKT + AT-DKT)

The behavioral vector z_t integrates the multi-KC item embedding with the three FoLiBiKT forgetting features and the elapsed time Δ_t:

$$z_t = g_\eta\Big([\, e_t^{q,kc};\ \mathrm{MLP}_c(c_t);\ \mathrm{MLP}_n(n_t);\ \mathrm{MLP}_g(g_t);\ \phi(\Delta_t) \,] \Big),$$

where $\phi(\cdot)$ is a sinusoidal time encoding, $[\,\cdot\,;\,\cdot\,]$ denotes concatenation, and g_η is a GRU cell. The output has shape (B, T, d_h).

Module 3. Option-aware Response Encoder (CD-MTL)

The option encoder augments z_t with the option embedding and the correctness label:

$$u_t = h_\phi(\, [\, z_t;\ o_t;\ r_t \,] \,),$$

where h_ϕ is a single feed-forward layer. This encoder is what allows the option-prediction head to be conditioned on the same latent representation that drives the correctness head, which is essential for the multi-task learning strategy of CD-MTL.

Module 4. Bayesian Knowledge State Tracker (UKT)

The state is modelled as a mean-field Gaussian:

$$\theta_t \sim \mathcal{N}(\mu_t,\ \mathrm{diag}(\sigma_t^2)),$$

where the variational parameters are produced by an LSTM-based recognition network:

$$(\mu_t,\ \log \sigma_t^2) = \mathrm{LSTM}_{\mathrm{enc}}(z_t,\ \theta_{t-1}).$$

The prior on θ_t is set to $\mathcal{N}(\theta_{t-1},\ \tau^2 I)$ with learnable τ, and we train the recognition network with the standard ELBO objective, consisting of a reconstruction term (the BCE loss on r_t) and a KL divergence $\mathrm{KL}\big(\mathcal{N}(\mu_t, \sigma_t^2) \,\|\, \mathcal{N}(\theta_{t-1}, \tau^2 I)\big)$ with weight β_{KL}. Following UKT, σ_t^2 is interpreted as the epistemic uncertainty.

Module 5. Adversarial Robustness Module (AT-DKT)

Let $\mathcal{L}_{\mathrm{clean}}$ denote the joint loss on the clean input. We apply an FGM-style perturbation on the latent vector z_t:

$$g = \nabla_{z_t} \mathcal{L}_{\mathrm{clean}}(z_t), \qquad z_t^{\mathrm{adv}} = z_t + \epsilon \cdot \frac{g}{\|g\|_2 + 10^{-8}},$$

and train the model with the combined loss $\mathcal{L} = \mathcal{L}_{\mathrm{clean}} + \alpha \mathcal{L}_{\mathrm{adv}}(z_t^{\mathrm{adv}})$. Unlike AT-DKT, which perturbs the *raw* input embedding, FOQUA-KT perturbs the *latent* behavioral vector, so the perturbation has a clear semantic interpretation: it is a worst-case perturbation of the model's internal knowledge representation.

Module 6. Multi-Task Prediction Heads (CD-MTL + FoLiBiKT)

Four heads share the same backbone (Modules 1-5) but use independent output layers.

1. **Correctness head**: $p_t^{\mathrm{corr}} = \sigma(W_1 \mu_t + b_1)$, trained with binary cross-entropy.

2. **Option head**: $p_t^{\text{opt}} = \text{softmax}(W_2 u_t)$, trained with cross-entropy. This is the head introduced in CD-MTL.
3. **Forgetting triplet head**: $\hat{c}_t = \sigma(w_c^\top z_t)$, $\hat{n}_t = \sigma(w_n^\top z_t)$, $\hat{g}_t = \sigma(w_g^\top z_t)$, each trained with binary cross-entropy. This is the head introduced in FoLiBiKT.
4. **Item-difficulty auxiliary head** (optional): a regression head on the IRT parameters (δ_q, α_q) that minimises $\text{MSE}(\hat{p}_t, \alpha_q(1 - p_t) + \delta_q)$.

The total loss is a weighted sum:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{corr}} + \lambda_2 \mathcal{L}_{\text{opt}} + \lambda_3 (\mathcal{L}_c + \mathcal{L}_n + \mathcal{L}_g) + \lambda_4 \mathcal{L}_{\text{KL}} + \lambda_5 \mathcal{L}_{\text{adv}} + \lambda_6 \mathcal{L}_{\text{diff}},$$

with all $\lambda_i \geq 0$ tuned on a validation set.

Module 7. Uncertainty Disambiguator (UKT + contrastive)

The two uncertainty sources are:

$$u_{\text{epi},t} = \mathbb{E}_k[\sigma_{t,k}^2], \qquad u_{\text{ale},t} = H(p_t^{\text{corr}}),$$

where $H(\cdot)$ is the binary entropy. To enforce a clean separation we add a contrastive auxiliary loss that *encourages* the two scores to be uncorrelated:

$$\mathcal{L}_{\text{unc}} = \left| \, \rho(u_{\text{epi}}, \, u_{\text{ale}}) \, \right|^2,$$

where ρ denotes the empirical Pearson correlation across the batch. A 95% confidence interval on p_t^{corr} is reported as $p_t^{\text{corr}} \pm 1.96\sqrt{u_{\text{epi},t}/B + u_{\text{ale},t}/4}$.

Module 8. Concept-Prerequisite Reasoner and Relevance Graph (QIKT + DIMKT)

The concept-prerequisite adjacency matrix is learned with a Hebbian-style update rule applied at every gradient step:

$$\Delta A_{\text{prereq},ij} = \eta_{\text{hebb}} \cdot \mathbb{E}_t\big[\, \sigma(\mu_{t,i}) \cdot \sigma(\mu_{t,j}) - A_{\text{prereq},ij} \,\big],$$

followed by a hard threshold ρ that zeroes out weak edges. The matrix is reported together with the symmetric item-item relevance matrix R_{sim} defined by cosine similarity in the multi-KC embedding space:

$$R_{\text{sim},ij} = \frac{\langle e_i^{q,kc}, \, e_j^{q,kc} \rangle}{\|e_i^{q,kc}\|_2 \, \|e_j^{q,kc}\|_2}.$$

C. Workflow Details

A single forward pass of FOQUA-KT proceeds as follows.

1. **Input encoding**: read $(i_t, o_t, r_t, \Delta_t, c_t, n_t, g_t)$ from the preprocessor; look up the corresponding embeddings; build the time encoding $\phi(\Delta_t)$.
2. **Multi-KC encoding** (Module 1): produce $e_t^{q,kc}$ from e_t^q and the Q-Matrix.
3. **Forgetting-aware encoding** (Module 2): fuse $e_t^{q,kc}$ with $(c_t, n_t, g_t, \Delta_t)$ into z_t via a GRU cell.
4. **Option-aware encoding** (Module 3): augment z_t with (o_t, r_t) into u_t.
5. **Bayesian state update** (Module 4): update the variational parameters (μ_t, σ_t^2) and sample $\theta_t \sim \mathcal{N}(\mu_t, \sigma_t^2)$.
6. **Adversarial perturbation** (Module 5): compute the FGM gradient on the joint loss and form z_t^{adv}; re-run steps 4-5 with the perturbed input and accumulate $\mathcal{L}_{\text{adv}}$.
7. **Multi-task prediction** (Module 6): compute all four heads and the four corresponding losses.
8. **Uncertainty disambiguation** (Module 7): compute u_{epi} and u_{ale} and the contrastive loss.
9. **Concept-prerequisite update** (Module 8): run the Hebbian update and threshold to obtain A_{prereq}; compute R_{sim}.
10. **Output aggregation**: collect the five outputs, log the loss terms and (if validation improves) update the parameters.

D. Data Compatibility Schema

The following table specifies the data flow between modules. The notation $\mathbb{R}^{a \times b \times c}$ denotes a real tensor of shape (a, b, c); (B, T, d) denotes batch, time and feature dimensions.

Source	Destination	Variable	Type	Shape	Pedagogical / Technical Meaning
Input	Module 1	i_t	int	(B, T)	Item id, identifies the practice question
Embedding table	Module 1	e_t^q	float	(B, T, d_q)	Raw item embedding
Q-Matrix	Module 1	Q	int	(N_q, N_c)	Item-concept incidence
Module 1	Module 2	$e_t^{q,kc}$	float	(B, T, N_c, d_k)	Multi-KC item embedding
Input	Module 2	(c_t, n_t, g_t)	int	$(B, T, 3)$	Forgetting-behavior labels
Input	Module 2	Δ_t	float	(B, T)	Elapsed time
Module 2	Module 4	z_t	float	(B, T, d_h)	Behavioral latent state
Module 2	Module 5	z_t	float	(B, T, d_h)	Behavioral latent state (clean)
Module 5	Module 6	z_t^{adv}	float	(B, T, d_h)	Adversarially-perturbed state
Module 4	Module 6	μ_t	float	(B, T, N_c)	Knowledge-state mean
Module 4	Module 6	σ_t^2	float	(B, T, N_c)	Knowledge-state variance
Input	Module 3	o_t	int	(B, T)	Selected multiple-choice option
Module 3	Module 6	u_t	float	(B, T, d_o)	Option-augmented representation
IRT parameters	Module 6	(δ_q, α_q)	float	$(B, T, 2)$	Item difficulty and discrimination
Module 6	Output	p^{corr}	float	(B, T)	Correctness probability
Module 6	Output	p^{opt}	float	(B, T, N_o)	Option distribution
Module 6	Output	$(\hat{c}, \hat{n}, \hat{g})$	float	$(B, T, 3)$	Forgetting-behavior prediction
Module 4	Module 7	σ_t^2	float	(B, T, N_c)	Epistemic uncertainty
Module 6	Module 7	p^{corr}	float	(B, T)	Aleatoric uncertainty (entropy)
Module 7	Output	$(u_{\mathrm{epi}}, u_{\mathrm{ale}})$	float	$(B, T, 2)$	Disambiguated uncertainties
Module 4	Output	μ_t	float	(B, T, N_c)	Per-concept mastery
Module 8	Output	A_{prereq}	float	(N_c, N_c)	Concept-prerequisite adjacency
Module 1	Module 8	$e^{q,kc}$	float	(B, T, N_c, d_k)	Item-concept relevance source
Module 8	Output	R_{sim}	float	(B, T_q, T_q)	Item-item relevance sub-graph

V. Conclusion

FOQUA-KT is a multi-task, uncertainty-aware, and adversarially-robust Knowledge Tracing architecture that consolidates six open-source KT papers into a single coherent design. The four mandatory criteria are addressed as follows. (1) Explainability is achieved through four prediction heads (correctness, option, forgetting behavior, difficulty), each of which is tied to a pedagogically meaningful quantity, and through a learned concept-prerequisite heatmap. (2) Uncertainty is explicitly decomposed into an epistemic component (the variational variance) and an aleatoric component (the predictive entropy), with a contrastive auxiliary loss that encourages the two to be uncorrelated.

(3) Mastery is represented as a per-concept variational Gaussian and is updated by a Bayesian recognition network, and a learned item-item relevance matrix is produced by cosine similarity in the multi-KC embedding space. (4) Data compatibility is specified in the schema of Section IV.D, with explicit tensor shapes and pedagogical annotations for every data flow.

We expect the proposed model to provide competitive AUC on ASSISTments 2009/2012, EdNet-KT1 and Eedi 2020, to outperform single-task baselines on the option-prediction metric, to yield well-calibrated uncertainty estimates as measured by expected calibration error, and to be more robust to input perturbations than non-adversarially-trained baselines.

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